

Digital Resilience in the Age of AI

When Critical Infrastructures Depend on Each Other

Stefan Schmid

Weizenbaum Institute & TU Berlin

Acknowledgements:

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A timely topic!

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NFP «Digitale Resilienz»

Die Digitalisierung unserer Gesellschaft bringt zahlreiche Vorteile mit sich, birgt aber auch Risiken. Cyberangriffe auf kritische Infrastrukturen nehmen zu: Spitäler kamen zum Stillstand, Ransomware-Angriffe beeinträchtigten den öffentlichen Verkehr und in öffentlichen Verwaltungen kam es zu Datenschutzverletzungen. Die Schweiz steht aufgrund ihrer besonderen Merkmale vor ganz eigenen Herausforderungen und Chancen. Das Land ist klein und weist einen hohen technologischen Entwicklungsstand auf.

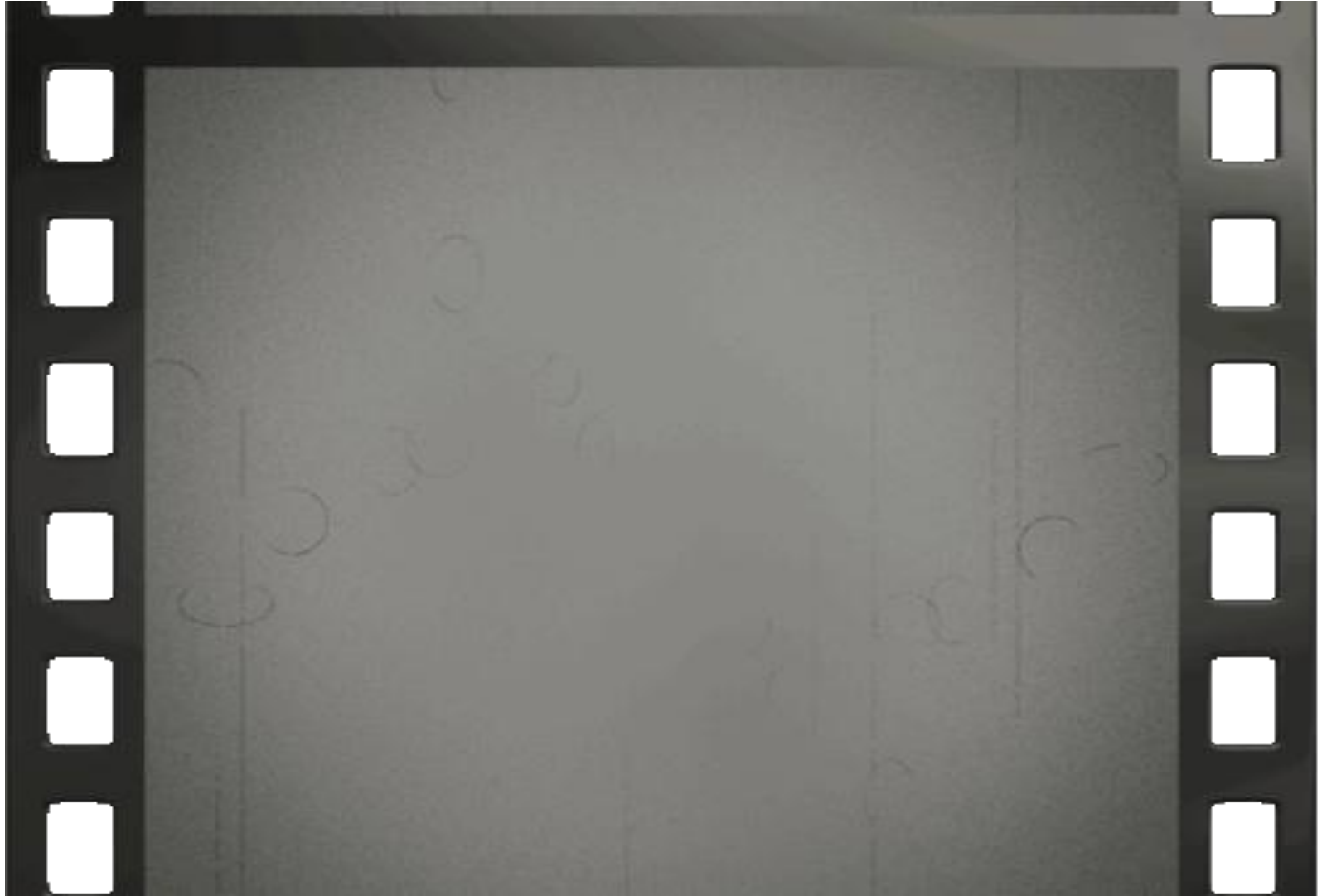
Das übergeordnete Ziel des Nationalen Forschungsprogramms «Digitale Resilienz kritischer Infrastrukturen» (NFP B6) ist es, die Widerstandsfähigkeit der Schweiz gegenüber Cyberangriffen langfristig zu stärken, indem eine agile und technologisch fortschrittliche kollektive Verteidigung entwickelt wird. Konkret geht es darum, die Fähigkeit kritischer Infrastrukturen zu verbessern, digitalen Störungen standzuhalten, sich anzupassen und sich davon zu erholen. Das Programm behandelt sowohl technische als auch praktische Aspekte, indem die Sozial- und Geisteswissenschaften in das Ökosystem der Cybersicherheit integriert werden.

Aktuell Forschung Förderung erhalten Über uns

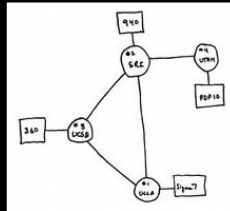
How did we get here?

Acknowledgements:

Winding back the time...



The Internet 50 Years Ago



- *Much smaller, for specific users: researchers*
- *Connectivity between fixed locations / “super computers”*
- *Simple applications like email and file transfer*

Kudos to: Pedro Casas

The Internet Evolution

A Huge Success Story

Today:

- Supports connectivity between *diverse “users”*: humans, machines, *things*, *AI bots*, ...
- Also supports wireless and *mobile* endpoints
- *Heterogeneous applications*: e-commerce, Internet telephony, VoD, gaming, etc.
- “One of the complex artefacts created by mankind” (Christos H. Papadimitriou)



Yet:

- Technology *hardly changed*! But now: mission-*critical infrastructure*

How Secure Are Networks?



The Internet at first sight:

- Monumental
- Passed the “Test-of-Time”
- Should not and cannot be changed

How Secure Are Networks?



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The Internet at second sight:

- Antique
- Brittle
- More and more successful attacks

Slide credit: Adrian Perrig

How Secure Are Networks?



The Internet at first sight:

- Monumental
- Passed the “Test-of-Time”
- Should not and cannot be changed



The Internet at second sight:

- Antique
- Brittle
- More and more successful attacks
- New threat model, f.ex.
quantum computation, super-intelligent AI bots, ...

Slide credit: Adrian Perrig

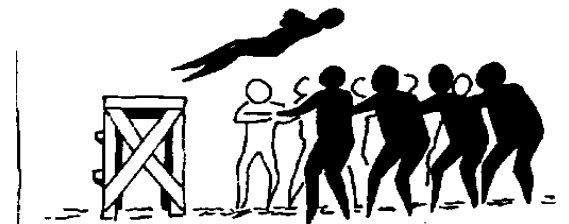
Why not secure?

Trust Assumptions

→ Internet in 80s: based on *trust*

→ Danny Hillis, TED talk, 2013:

“There were two Dannys. *I knew both*. Not everyone knew everyone, but there was an atmosphere of trust.”



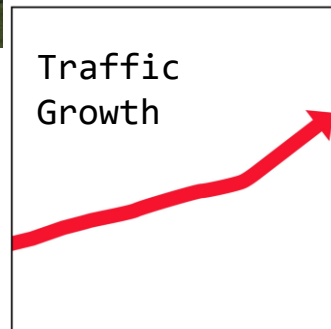
With cloud computing and AI: Datacenters also a Critical Infrastructure



Datacenters (“hyper-scale”)



Interconnecting networks:
a **critical infrastructure**
of our digital society.



Source: Facebook

With cloud computing and AI: Datacenters also a Critical Infrastructure



Datacenters (“hyper-scale”)



+network

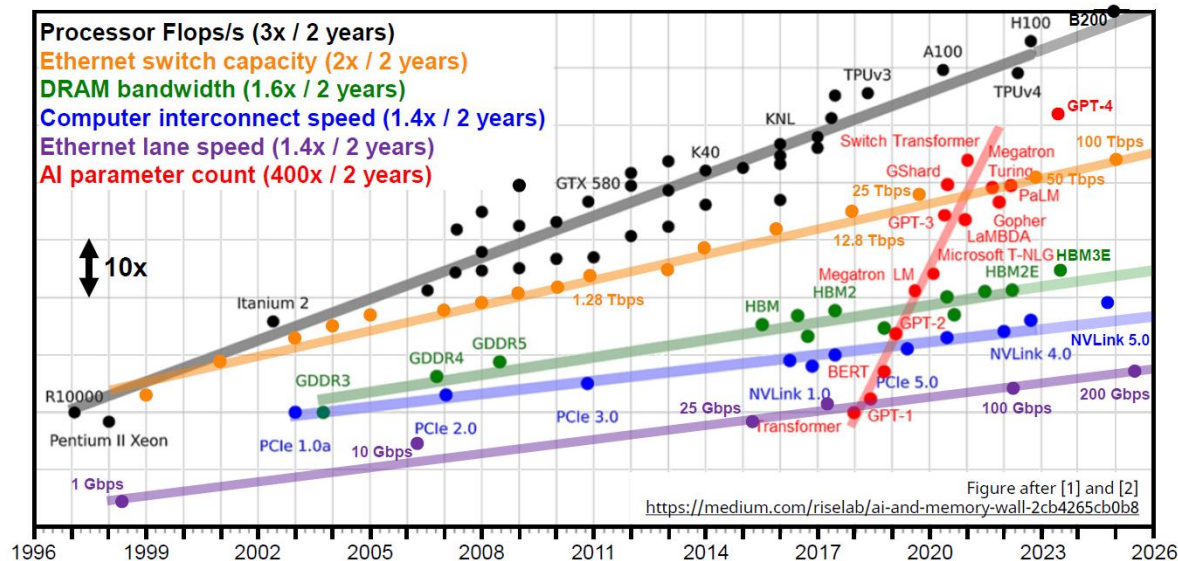
Interconnecting networks:
a **critical infrastructure**
of our digital society.



Credits: Marco Chiesa

Technological trends

Network Size Increases



Credits: Nicola Calabretta

- Networks: critical but also larger and larger (*more equipment*)
- Increased *attack surface*?

Complexity: leads to unintended outages

The Human in the Loop

Countries disconnected

Data Centre • Networks

Google routing blunder sent Japan's Internet dark on Friday

Another big BGP blunder

By Richard Chirgwin 27 Aug 2017 at 22:35

40 SHARE ▼

Last Friday, someone in Google fat-thumb'd a border gateway protocol (BGP) advertisement and sent Japanese Internet traffic into a black hole.

The trouble began when The Chocolate Factory "leaked" a big route table to Verizon, the result of which was traffic from Japanese giants like NTT and KDDI was sent to Google on the expectation it would be treated as transit.

Passengers stranded

British Airways' latest Total Inability To Support Upwardness of Planes* caused by Amadeus system outage

Stuck on the ground awaiting a load sheet? Here's why

By Gareth Corfield 19 Jul 2018 at 11:16

109 SHARE ▼



BA flights around the world were grounded as a result of the Amadeus outage

Even 911 affected

Officials: Human error to blame in Minn. 911 outage

According to a press release, CenturyLink told department of public safety that human error by an employee of a third party vendor was to blame for the outage

Aug 16, 2018

Duluth News Tribune

SAINT PAUL, Minn. — The Minnesota Department of Public Safety Emergency Communication Networks division was told by its 911 provider that an Aug. 1 outage was caused by human error.

Even tech-savvy companies struggle:

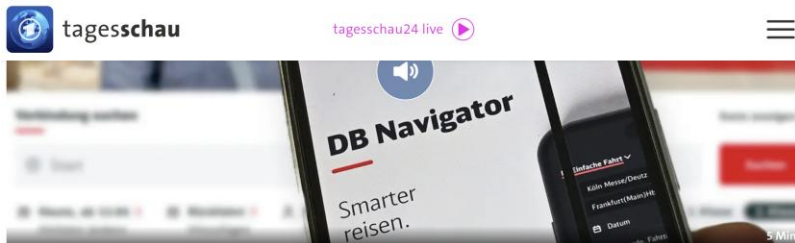


Mainly:
human
errors!

Slide credits: Nate Foster and Laurent Vanbever

Threats in the news

(D)DoS Attacks



Cyberangriff auf die Bahn

BSI geht nicht von weiteren Auswirkungen aus

Stand: 19.02.2026 • 15:05 Uhr

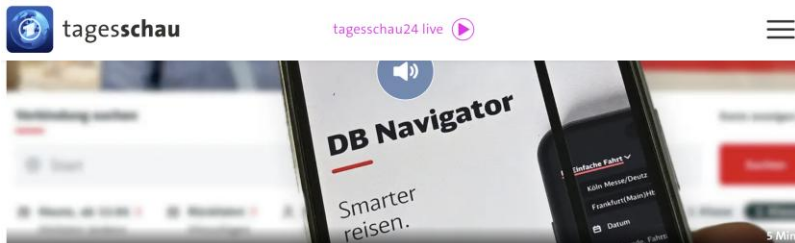
Nach dem Cyberangriff auf die Bahn ist noch unklar, wer dafür verantwortlich ist. Laut dem Bundesamt für Sicherheit in der Informationstechnik gibt es keine weiteren Auswirkungen.

Wer hinter dem Cyberangriff auf die Bahn steckt, ist noch unklar - von dem Angriff waren Auskunfts- und Buchungssysteme am Dienstag und Mittwoch betroffen. Nach Angaben des Bundesamts für Sicherheit in der Informationstechnik (BSI) hat der Cyberangriff keine weiteren Auswirkungen.

„Die **Bahn** hatte am Dienstagnachmittag erstmals über die Störungen der Auskunfts- und Buchungssysteme auf der Website und in der Navigator-App informiert. Es handelt sich um eine sogenannte **DDoS-Attacke** (Distributed Denial of Service), eine digitale Überlastungsattacke.“

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- Large attacks using **IoT devices** („babyphone attack“).
- Banks paying **extortion** money?

Remark 1 on (D)DoS

Botnets and Blockchain

- **Botnets** (that perform DDoS) have **advanced architectures**
- F.ex., botnets use **blockchains** to eliminate centralized **command-and-control** servers and resist traditional law enforcement takedowns

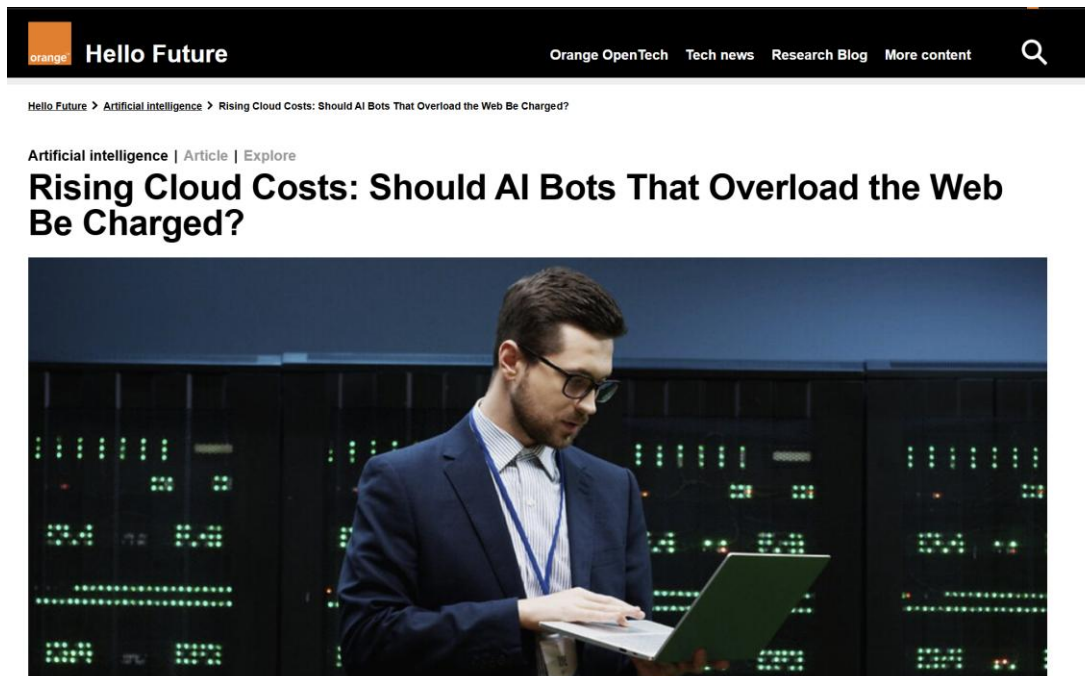
[Main](#) — [Library](#) — [Blog](#) — Exploring Aeternum C2: a new botnet that lives on the blockchain

Exploring Aeternum C2: a new botnet that lives on the blockchain

Remark 2 on (D)DoS

AI Bots: Resource Consumption

*Remark, even if not malicious:
much (soon: most?) Internet traffic by bots?*



Threats also on higher layers

Phishing Attacks

Spionage/phishing Bundestags Abgeordnete via Messenger.

≡ Menü

DIE  ZEIT

Abo testen



Phishing-Angriffe auf Signal

Warum Politiker derzeit so viel Angst vor Spionage haben wie noch nie

Seit Monaten verschicken Unbekannte Phishing-Mails über den Chat-Dienst Signal. Betroffen sind auch hochrangige Politikerinnen wie Julia Klöckner. Wer steckt dahinter?

Von [Dune Korth](#) und [Ingo Malcher](#)

Aktualisiert am 24. April 2026, 15:24 Uhr ⓘ

Threats also on higher layers

Phishing Attacks

Solution?!

☰ 🔍 **Golem** [Also testen](#) [Anmelden](#)

NACH SIGNAL-PHISHING

Bundestag forciert Umstieg auf Wire

Seit 2020 testen Bundesbehörden den Messenger Wire. Nun könnte der Deutsche Bundestag diesen nutzen und Dienste wie Signal untersagen.

28. April 2026 um 08:05 Uhr / [Friedhelm Greis](#)

14 [News folgen](#) [Teilen](#)



Bild: Sven Gellert/Getty Images

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Sicherheit in der digitalen Kommunikation

Bundestag plant Umstieg von Signal auf Wire

27. April 2026, 15:06 Uhr | Lesezeit: 3 Min. | [20 Kommentare](#)



Worüber die Bundestagsabgeordneten auf ihren eigenen Smartphones kommunizieren, ist ihre Entscheidung. Doch um die Sicherheit zu erhöhen, will das Parlament auf den europäischen Anbieter Wire umsteigen.

Helena Dolderer/picture alliance/dpa

Threats also on higher layers

Ransomware

heise online > IT > Cyberangriff: TU Berlin rechnet mit monatelangen IT-Einschränkungen

Cyberangriff: TU Berlin rechnet mit monatelangen IT-Einschränkungen

Es wird noch einige Zeit dauern, bis die zentralen IT-Systeme der TU Berlin nach der Ransomware-Attacke wieder laufen. Auch das SAP-Kernsystem ist betroffen.

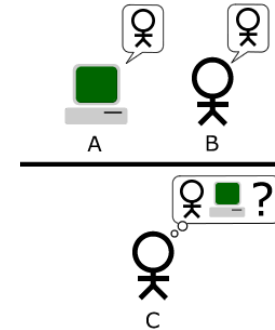
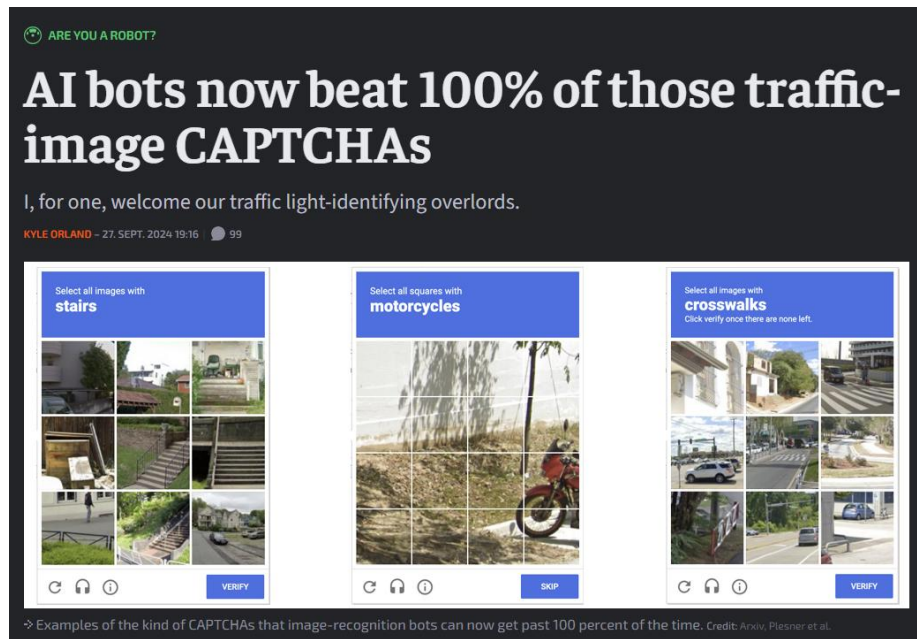
🔊 🖨️ 💬 298



Example at *TU Berlin*

Threats related to AI

AI Bots: CAPTCHAs Broken?

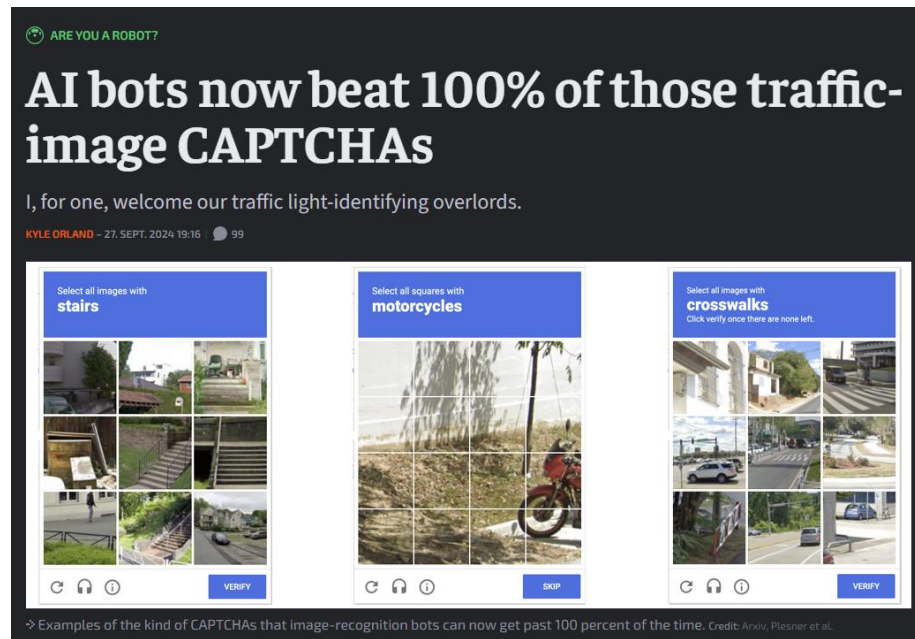


*Turing Test:
(wo)man or machine?*

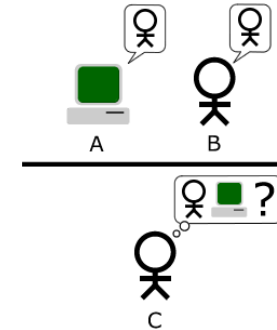
*Particularly relevant in context
of Weizenbaum/ELIZA.*

Threats related to AI

AI Bots: CAPTCHAs Broken?



Particularly relevant in context of Weizenbaum/ELIZA.



*Turing Test:
(wo)man or machine?*



Will AI bots exploit us?

Threats related to AI

(AI) Biases and Steering

Elon Musk says he'll create 'TruthGPT' to counter AI 'bias' 

April 2023

When ChatGPT got too friendly—why OpenAI rolled back its April update

"The model noticeably became more sycophantic," OpenAI admitted in a detailed post. "It aimed to please the user, not just as flattery, but also as validating doubts, fueling anger, urging impulsive actions, or reinforcing negative emotions in ways that were not intended."

ChatGPT: This is how green the worlds most famous AI thinks



May 2024



→ *Political biases?*

GermanPartiesQA: Benchmarking Commercial Large Language Models and AI Companions for Political Alignment and Sycophancy

Jan Batzner^{W, TUM}, Volker Stocker^{W, TUB}, Stefan Schmid^{W, TUB}, Gjergji Kasneci^{TUM}

^WWeizenbaum Institute ^{TUB}Technical University Berlin ^{TUM}Technical University Munich

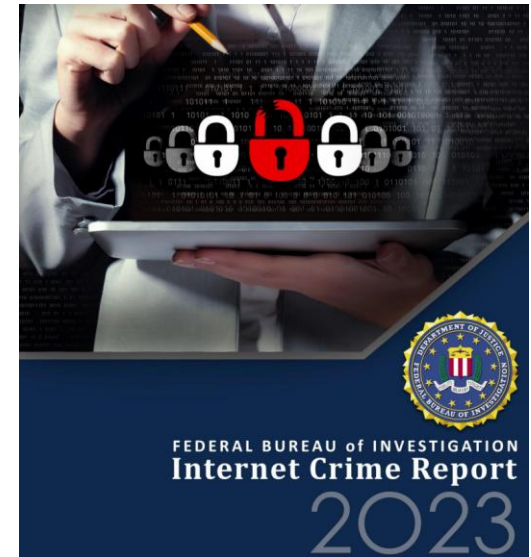
Threats related to AI

(AI) Biases and Steering

Urgent calls for *regulation* to establish guardrails.

Example: FBI's 2023 Internet Crime Report shows that *social engineering* led to financial losses >\$12.5 billion

And EU Agency for Cybersecurity's 2023 report shows that *synthetic media* is hazardous to *digital identity* and trust



More threats

Security&Privacy-Awareness!?

→ Privacy *awareness* of citizens:

police/governments vs *social media* vs big *companies* (Google)

→ Case study Twitter/X and Bitcoin

Pseudonymity at Risk: Linkage Attacks on Blockchain Users with Off-Chain Cues

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San Francisco, USA
friedhelm@trmlabs.com

Threats on other type of networks

Critical Infrastructure

Brandanschlag auf das Berliner Stromnetz 2026

Koordinaten: 52° 25' 32" N, 13° 18' 26" O | 

 2 Sprachen ▾

Artikel Diskussion

Lesen Bearbeiten Quelltext bearbeiten Versionsgeschichte ⋮

Am Morgen des 3. Januar 2026, einem Samstag, wurde ein **Brandanschlag auf das Berliner Stromnetz** verübt, der einen flächenhaften **Stromausfall** für etwa 45.000 Haushalte und über 2.200 Betriebe im Südwesten **Berlins** zur Folge hatte. Der **Anschlag** wurde durch **Brandstiftung** an einer Kabelbrücke in **Berlin-Lichterfelde** ausgeführt und sorgte für den längsten Stromausfall in der Stadt seit dem **Kriegsende** 1945.^[1] Bei tiefwinterlichem Wetter kam es so auch zum Ausfall vieler Heizungen. Am 7. Januar 2026 konnten die Folgen vollständig behoben werden.^{[2][3]}

Im Internet wurde ein **Bekennerschreiben** zur Tat im Namen einer **linksextremen Vulkangruppe** veröffentlicht. Wenige Tage später distanzierte sich jedoch eine angeblich ursprüngliche Vulkangruppe von den in ihrem Namen durchgeführten Aktionen der letzten Jahre.^[4]

Am 6. Januar 2026 übernahm der **Generalbundesanwalt beim Bundesgerichtshof** die Ermittlungen, weil den Tätern unter anderem die **Mitgliedschaft in einer terroristischen Vereinigung** vorgeworfen wird. Das mangelhafte Krisenmanagement des Regierenden Bürgermeisters **Kai Wegner** (CDU) führte im Juli 2026 zu dessen Rückzug als erneuter Kandidat für das Amt.^[5]



Die vom Brandanschlag betroffene Kabelbrücke, dahinter das Kraftwerk Lichterfelde (2009)

Threats on other type of networks

Critical Infrastructure

- Worrisome also because networks are *interdependent*
- E.g. *power* outage also affects *communication* network (mobile phone still works but what about *celltower*, hotspot, routers, etc.?)
- Goal of Berlin senate: communication for at least 12 more hours (but requires more *batteries*, more *maintenance*, more *space*...: *cost* intensive!)

rbb 24

Wahl Berlin Brandenburg Politik Wirtschaft Kultur Sport Panorama Verkehr

Interview Mobilfunk bei Stromausfall

"Viele dieser Netze sind nicht dazu ausgelegt, im Krisenfall länger zu funktionieren"

Di 03.02.2026, 08:17 Uhr



Audio: rbb24 | 03.02.2026 | Annette Kufner

What can we do?!

For example, a question of BSI: how to build secure datacenter networks for the Government?

What can we do?!

For example, a question of BSI: how to build secure datacenter networks for the Government? ... if it is too expensive to build own (trusted) hardware and hence needs to be bought externally.

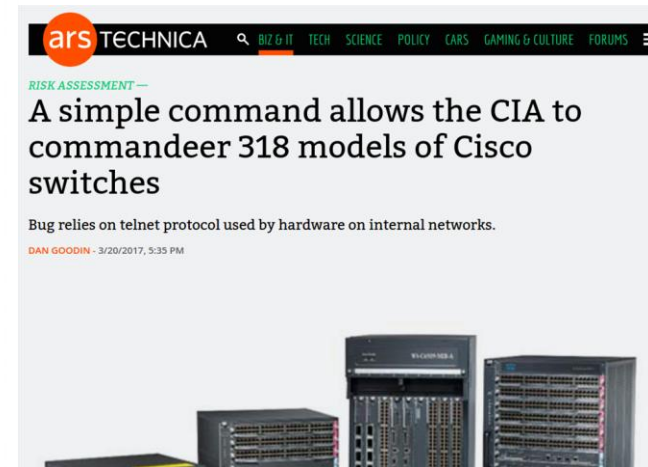
Fundamental Question

Can we trust technology?

(TS//SI//NF) Such operations involving **supply-chain interdiction** are some of the most productive operations in TAO, because they pre-position access points into hard target networks around the world.



(TS//SI//NF) Left: Intercepted packages are opened carefully; Right: A “load station” implants a beacon



- **Hardware backdoors** and exploits
- Fundamental question: how can we **hope to build a secure network** if the underlying hardware can be insecure?!
- E.g., **secure cloud for the government**: no resources and expertise to build own “trustworthy” high-speed hardware

Fundamental Question

Can we trust companies?



February 2020: For more than half a century, *governments all over the world* trusted a single company to keep the communications of their spies, soldiers and diplomats secret. But: Crypto AG was *secretly owned by the CIA!?*

How to protect us?

Creative Solutions



The New York Times



Activate This 'Bracelet of Silence,' and Alexa Can't Eavesdrop

Microphones and cameras lurk everywhere. You may want to slip on some privacy armor.



February 2020: Wearable microphone jamming.
(<https://www.mirror.co.uk/tech/alexa-owners-can-stop-eavesdropping-21539032>)

How to protect us?

Creative Solutions



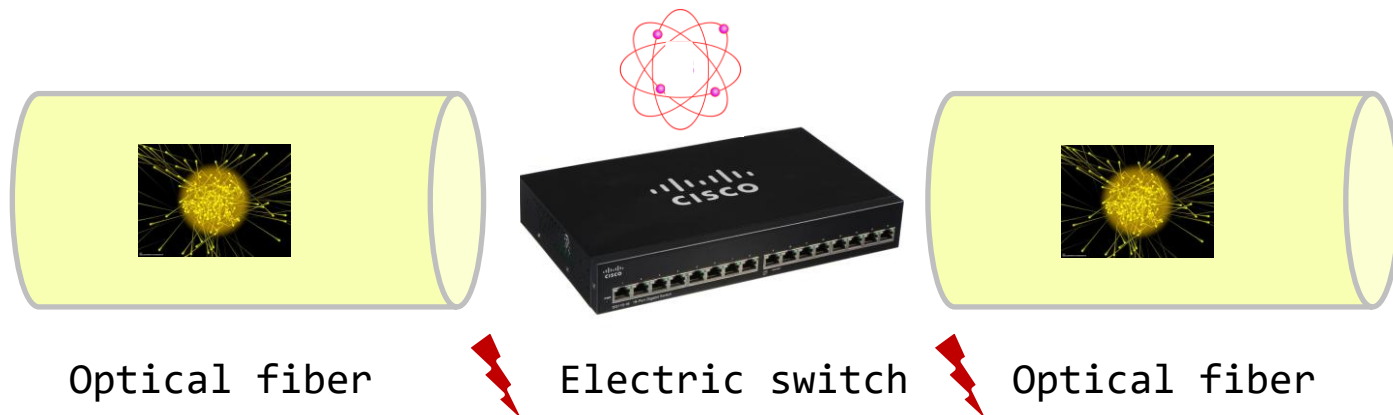
Glasses developed by Scott Urban *reflect infrared light* from security cameras to blur out the wearer's face.

How to protect us?

Optical Technologies

→ **Electronic** devices (e.g., routers)

→ Issues: **conversion** photon-electron-photon (**bottleneck** and **energy** inefficient), but also **not robust** to interference, **water**, etc.

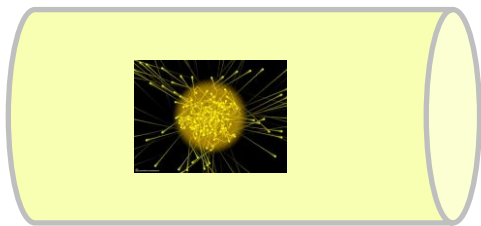


How to protect us?

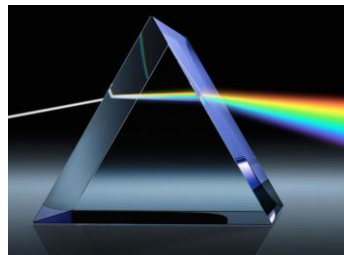
Optical Technologies

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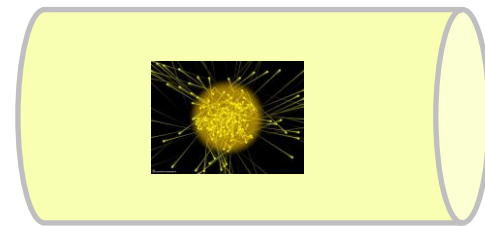
→ Issues: **conversion** photon-electron-photon (**bottleneck** and **energy** inefficient), but also **not robust** to interference, **water**, etc.



Optical fiber



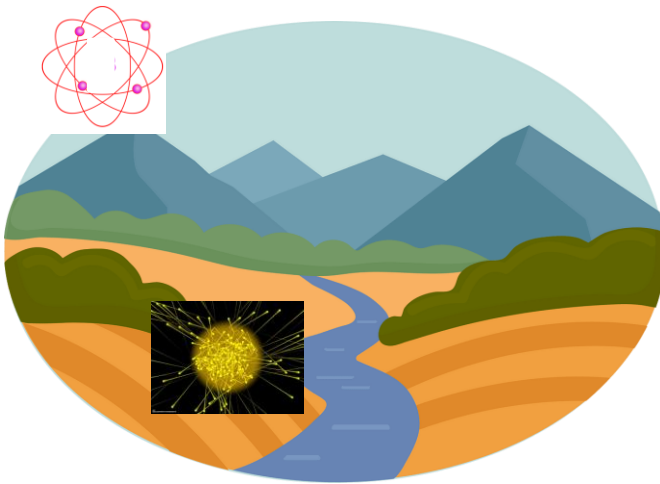
Optical switch



Optical fiber

Optical Networks: Resilience

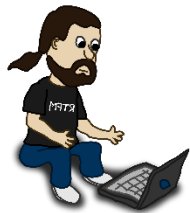
Floodings in South Germany destroyed much electrical network infrastructure



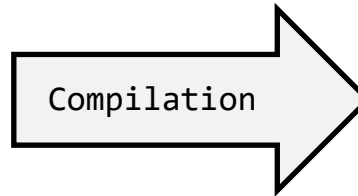
Solution: deploy optical infrastructure (in valleys) and electrical **on hills** where safe?

How to protect us?

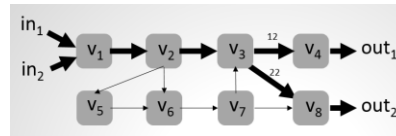
Formal Methods



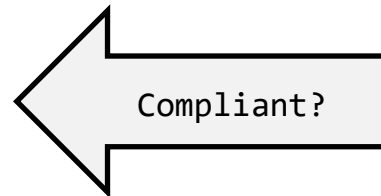
FT	In-I	In-Label	Out-I	op
τ_{v_1}	in_1	$\perp$	(v_1, v_2)	$push(10)$
	in_2	$\perp$	(v_1, v_2)	$push(20)$
τ_{v_2}	(v_1, v_2)	10	(v_2, v_3)	$swap(11)$
	(v_1, v_2)	20	(v_2, v_3)	$swap(21)$
τ_{v_3}	(v_2, v_3)	11	(v_3, v_4)	$swap(12)$
	(v_2, v_3)	21	(v_3, v_4)	$swap(22)$
τ_{v_4}	(v_2, v_3)	11	(v_3, v_4)	$swap(12)$
	(v_2, v_3)	21	(v_3, v_4)	$swap(22)$
τ_{v_5}	(v_3, v_4)	12	out_1	pop
τ_{v_6}	(v_2, v_3)	40	(v_5, v_6)	pop
τ_{v_7}	(v_2, v_3)	30	(v_6, v_7)	$swap(31)$
	(v_5, v_6)	30	(v_6, v_7)	$swap(31)$
τ_{v_8}	(v_5, v_6)	61	(v_6, v_7)	$swap(62)$
	(v_5, v_6)	71	(v_6, v_7)	$swap(72)$
τ_{v_9}	(v_6, v_7)	31	(v_7, v_8)	pop
	(v_6, v_7)	62	(v_7, v_8)	$swap(11)$
$\tau_{v_{10}}$	(v_6, v_7)	72	(v_7, v_8)	$swap(22)$
	(v_3, v_4)	22	out_2	pop
$\tau_{v_{11}}$	(v_7, v_8)	22	out_2	pop



$pX \Rightarrow qXX$
 $pX \Rightarrow qYX$
 $qY \Rightarrow rYY$
 $rY \Rightarrow r$
 $rX \Rightarrow pX$



local FFT	Out-I	In-Label	Out-I	op
τ_{v_2}	(v_2, v_3)	11	(v_2, v_6)	$push(30)$
	(v_2, v_3)	21	(v_2, v_6)	$push(30)$
	(v_2, v_6)	30	(v_2, v_5)	$push(40)$
global FFT	Out-I	In-Label	Out-I	op
τ_{v_2}	(v_2, v_3)	11	(v_2, v_6)	$swap(61)$
	(v_2, v_3)	21	(v_2, v_6)	$swap(71)$
	(v_2, v_6)	61	(v_2, v_5)	$push(40)$
	(v_2, v_6)	71	(v_2, v_5)	$push(40)$



Formal language
 which supports
automated analysis

How to protect us?

Like Human: Fast and Slow

Two complementary mechanisms for resilience

Fast local reactions to keep things running – and slower, deliberate decisions to improve the system.

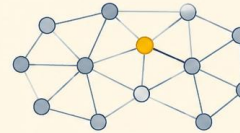
Brain

Global view. Deliberate decisions.

Analyzes the situation, finds better solutions, and adapts the system in the longer term.

Slow and intelligent adaptation

Network-wide optimization and reconfiguration



- Global network view
- Computes improved routes
- Reconfigures the network
- Learns from experience
- Operates on seconds, minutes or longer

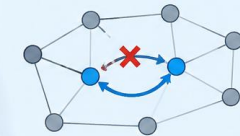
Spinal cord

Fast local reaction.

Reacts immediately to disruptions using simple, local rules – keeping the system running.

Fast and automatic protection

Local reaction (e.g., fast reroute)



- Local information only
- Immediate reaction (ms)
- Pre-computed or simple rules
- Keeps traffic flowing
- Works without waiting for the global controller

Together, they make the system resilient.

Immediate stability – and continuous improvement.

AI generated...

How to protect us?

Regulations, Market (and User Awareness)

„AI Act and DSA provide important **minimal requirements**, especially regarding illegal contents.“

„We need a comprehensive approach combining value chain **regulation**, content **moderation**, **competition**, and ongoing **technology** design governance.“

„We also need more **transparency** on training data and algorithms (e.g. for ranking).“

„We need an **economic market** where there are different and independent LLM providers (like there are different newspapers).“



How to protect us?

Regulations, Market (and User Awareness)

Communications of the ACM

Research and Advances

Artificial Intelligence and Machine Learning

Communication Bias in Large Language Models: A Regulatory Perspective

Though new regulatory frameworks address fairness, accountability, and safety in AI systems, they often fail to directly mitigate the subtle communication bias in LLMs that can distort public discourse and democratic processes.

By Adrian Kuenzler and Stefan Schmid

Posted Mar 13 2026

Artificial intelligence (AI) involves computational systems designed to perform tasks that traditionally require human intelligence, such as perception, reasoning, and language understanding. Large language models (LLMs), a prominent subset of AI, are built on advanced neural network architectures that can generate new data, including text, images, and audio. LLMs use various technologies to identify patterns in a given set of training data, without requiring explicit instructions about what to look for.^{12,35} LLMs typically assume that the training data follows a probability distribution; once they have identified existing patterns, they can generate new instances that are similar to the original data. By drawing from and combining training data, LLMs can create new content that transcends the initial dataset.¹⁷ But this process has technical limitations. For instance, LLMs can hallucinate by producing plausible yet factually incorrect or fabricated information; they can struggle with ambiguous or missing data, leading to blind spots in their outputs; and they can reflect and amplify existing preconceptions present in their algorithms or training data, leading to *communication bias*, which we focus on here.

Feedback



How to protect us?

Measure, Monitor, Benchmark...

Biases in LLMs:

GermanPartiesQA: Benchmarking Commercial Large Language Models and AI Companions for Political Alignment and Sycophancy

Jan Batzner^{W, TUM}, Volker Stocker^{W, TUB}, Stefan Schmid^{W, TUB}, Gjergji Kasneci^{TUM}

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Robustness of satellite networks:

Multi-Modal Measurements of Weather Effects and Link Quality Dynamics in LEO Satellite Networks

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¹Technical University of Berlin ²Technical University of Munich ³Fraunhofer SIT ⁴Weizenbaum Institute

Blockchains networks:

**Friend or Foe?
Identifying Anomalous Peers in Monero's P2P Network**

Yannik Kopyciok
TU Berlin

Stefan Schmid
TU Berlin and Fraunhofer SIT

Friedhelm Victor
TRM Labs

Thank you! Questions?